

Module 06: Lab 06

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Exercises

- 1) **Why do the two RMSE plots show very different behavior? Use underfitting and overfitting in your answer, as well as the bias-variance tradeoff.**

The two RMSE plots show very different behavior due to the concepts of underfitting, overfitting, and the bias-variance trade-off.

- **First RMSE Plot:**
 - This plot shows that as the degree of polynomial increases the RMSE decreases. This is most likely due to the fact that as the model becomes more complex and starts to fit the training data better and reduces the error.
 - However a higher degree polynomial may lead of overfitting our data and on top of showing relationships between our variables also show the noise in the training data, this the lower RMSE value.
 - Overfitting is characterized by low bias but high variance. The model is too complex and fits the training data too closely, including the noise.
- **Second RMSE Plot:**
 - This plot shows a U-shaped curve where RMSE decreases as the degree of polynomial increas but then also increases as after a certain degree of the polynomial. This is because while a more complex model fits the training data well, it may not generalize well to new data.
 - Lower-degree polynomials underfit the data, which means they have high bias but low variance. They are too simple to capture the true underlying relationship.

- 2) **Using the last plot above, what degree of polynomial should we choose? After running all of the above code, substitute the degree you chose for the question mark in the code below and run it. Copy and paste the resulting plot.**

After we run all our codes, it seems like the degree best for not overfitting and underfitting would be between 2 and 4 (from our two graphs) so we are choosing 4.

